

HTML



UX

SEO

HTML 5

CMS



Infomax.org.in

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Objective:

After completing this unit, Learner will be able to :

- Develop static website using different HTML Controls.
- Able to Create interactive Form to accept data from user.
- Able to represent data/information in different layouts and formats.

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Introduction to HTML

HTML (HyperText Markup Language) is used to create web documents known as **web pages**. Every website you see on the internet is built using HTML in one form or another.

HTML is **not a programming language** because it does not perform calculations or logic. Instead, it is a **markup language**, which means it is used to **structure and format content** on a web page.

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HTML works using a collection of special keywords called **tags**. These tags tell the browser how to display text, images, links, and other elements.

HTML is **not case-sensitive**, which means tags can be written in **uppercase or lowercase**.

For example:

`<HTML>` and `<html>` both work the same.

HTML was created by **Tim Berners-Lee** in **1991**, but it was officially released in **1995** as **HTML 2.0**.

The **latest version** of HTML used today is **HTML5**, which supports multimedia, graphics, and modern web features.

HTML Tags

HTML tags are instructions given to the web browser to display content in a specific way.

Tags are the **basic building blocks of HTML**. They define how text appears on the screen, such as headings, paragraphs, images, and links.

All HTML tags:

- Start with an **opening angle bracket** <
- End with a **closing angle bracket** >

Types of HTML Tags

HTML tags are mainly of two types:

- Paired Tags
- Empty (Singular) Tags

Paired Tag

Paired tags have both:

An **opening tag**

A **closing tag**

They are also called **container tags** because they hold text or other HTML tags inside them.

Examples of Paired Tags

```
<body></body>
```

```
<p>sfgsfsdf</p>
```

```
<b></b>
```

Example:

```
<p>This is a paragraph.</p>
```

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Empty (Singular) Tags

Empty tags have **only an opening tag** and **no closing tag**. They do not contain any content. These tags are also known as **singular tags**.

Examples of Empty Tags

`<br />` – line break

`<hr />` – horizontal line

`<img />` – image

Example: `<br />`

Structure of HTML page

Every HTML page follows a basic structure that helps the browser understand the document.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

Some tags are placed here

```
</head>
```

```
<body>
```

Some text or tags are placed here

```
</body>
```

```
</html>
```

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Explanation:

- `<!DOCTYPE html>` tells the browser that the document is HTML5.
- `<html>` is the root element.
- `<head>` contains information about the page.
- `<body>` contains visible content shown on the screen.

HTML Tag <HTML>

The <html> tag:

Encloses **all other HTML tags**

Acts as the **root container** for the web page

Although the <html> tag is technically **optional**, it is always considered a **good practice** to include it so the browser can clearly understand the document structure.

Note: **Web browsers follow HTML parsing rules**", it means:

- ☐ Browsers read your HTML code using fixed rules.
- ☐ If they find mistakes, they try to correct them automatically.

Head Section and Elements of Head Section

The <head> section:

Is placed **inside the <html> tag**

Contains information **about** the webpage, not visible content

Helps browsers, search engines, and tabs understand the page

The <head> section may include:

<title>

Metadata <meta>

Stylesheets <style></style>

Scripts <script></script>



Elements of Head Section

The <title> Tag

The <title> tag:

Defines the **title** of the web page

Appears on the **browser tab**

Can be used **only once** in a document

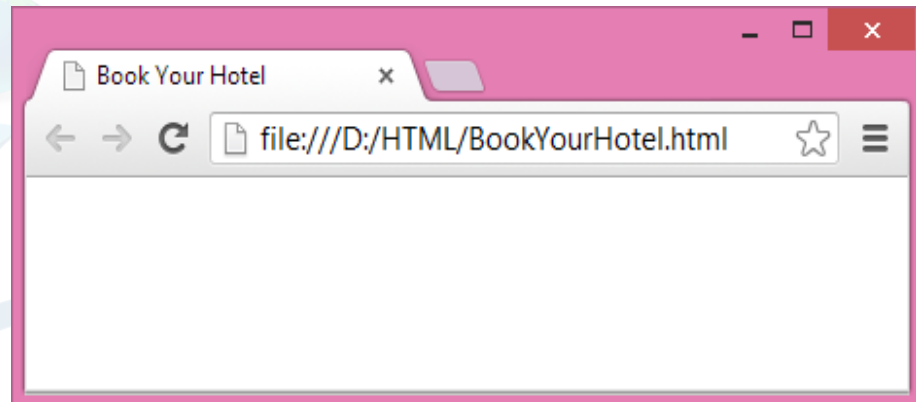
Example:

```
<head>
```

```
  <title>Book Your Hotel</title>
```

```
</head>
```

Note: If you write plain text inside <head>, the browser may move it to <body> and display it.



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The body section

The **body section** is the most important part of an HTML document because it contains **everything that appears on the web page**.

It begins with the `<body>` tag.

It holds all the visible content such as text, images, links, tables, and videos.

Whatever you write inside the body tag is displayed in the browser window.

Example:

Text, headings, images, and links that you see on a webpage are all placed inside the `<body>` section.

Attribute of Tag

Attributes provide **additional information** about an HTML element. They help modify the appearance or behavior of elements.

Important points about attributes:

Attributes are written **inside the opening tag**.

Every attribute has **two parts**:

Name – what property you want to change

Value – how you want to change it

General Syntax –

```
<tagname attribute="value">content</tagname>
```

Example:

```
<p align="center">This is a paragraph</p>
```

Here:

- **align** is the attribute name
- **center** is the value
- It aligns the paragraph text to the center

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Attributes of Body Tag:

BGCOLOR

- To set the background color page (color name or color code)

TEXT

- To set the text color of web page (color name or color code)

TOPMARGIN

- To set the margin from top of page (px or %)

LEFTMARGIN

- To set the margin from left & right of page (px or %)

BACKGROUND

- To set the background image on web page (url/path)

LINK

- To set the link color of page (color name or color code)

VLINK

- It sets the color of a link **after you have already clicked it.**

ALINK

- alink is used to define the color of a link **while it is being clicked/active.**

ALIGN

To set the alignment of body content (left, right, center)

Formatting Tags:

Formatting tags help organize and style text on a webpage.

Heading Tags :-

Heading tags are used to define **titles and headings** on a webpage. They range from <h1> (largest) to <h6> (smallest).

<h1> is used for main headings

<h6> is used for smallest headings

Attribute of Heading Tag

align – aligns the heading (left, right, center)

Example:<h1 align="center">Welcome to My Website</h1>

Paragraph Tag :-

The <p> tag is used to define a paragraph.

Attribute of <p> Tag

align – left, right, center, justify

Example:<p align="justify">This paragraph explains the topic in detail.</p>

BGCOLOR



```
<html>  
<body bgcolor="yellow">  
This page has yellow background.  
</body>  
</html>
```

TEXT



```
<body text="blue">  
This text will appear blue.  
</body>
```

TOPMARGIN



```
<body topmargin="50">  
Content starts 50px from top.  
</body>
```

LEFTMARGIN



```
<body leftmargin="40">  
Content starts 40px from left.  
</body>
```

BACKGROUND → `<body background="image.jpg">
</body>`

LINK → `<body link="red">
Click Here
</body>`

VLINK → `<body link="blue" vlink="red"> <h2>HTML Link and VLink Example</h2>
<p> Visit Google </p>
<p>Visit Wikipedia </p></body>`

ALINK → `<body link="blue" vlink="red" alink="green"> <h2>HTML Link, VLink and
ALink Example</h2> <p> Visit
Google </p> <p> Visit
Wikipedia </p></body>`

ALIGN → `<body align="center">
This content is centered.
</body>`

Formatting Tags (Conti.)

Line Break

The `<br />` tag is used to move text to the **next line**.

It is an **empty (singular) tag**.

Example: Hello`<br />`World

Horizontal Line

The `<hr />` tag is used to draw a horizontal line across the page.

Attributes of `<hr />` Tag

color – sets line color

width – sets width (default 100%)

size – sets thickness

align – left, right, center

Example:

```
<hr color="red" width="50%" size="4" align="center">
```

Text Formatting Tags

- **Bold** : `<b></b>` or `<strong></strong>` Bold Text
- **Italic** : `<i></i>` Italic Text
- **Underline** : `<u></u>` Underline text.
- **Strikethrough** : `<strike></strike>` Line through on text
- **Superscript** : `<sup></sup>` Superscript
- **Subscript** : `<sub></sub>` Subscript.
- **Big** : `<big></big>` Big size Text
- **Small** : `<small></small>` small text
- **Mark** : `<mark></mark>` Highlight text.
- **Insert** : `<ins></ins>` insert text just like underline.
- **Pre** : `<pre></pre>` Preformatted text.

Formatting Tags (Conti.)

The tag is used to change the **color, size, and style** of text.

Attributes of Tag

color – sets text color

size – sets text size (1 to 7, default is 3)

face – sets font style (e.g., Times New Roman)

Example :

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Formatting Tags (Conti.)

Scrolling Text or Images

The <marquee> tag is used to create **scrolling text or images**, similar to news tickers on TV.

Attributes of <marquee> Tag

direction – left, right, up, down

scrollamount – controls speed

width – sets width of scroll area

height – sets height

behavior – scroll, slide, alternate

bgcolor – sets background color

Example :

```
<marquee direction="left" scrollamount="6" width="600" height="40"
bgcolor="yellow" behavior="scroll">
```

□ Breaking News: HTML is easy and fun to learn! □

```
</marquee>
```

HTML Entities

HTML entities are used to display reserved characters in HTML.

Symbol	Entity	Symbol	Entity
Space	 	← (LEFTWARDS ARROW)	←
< (Less than)	<	↑ (UPWARDS ARROW)	↑
> (Greater than)	>	→ (RIGHTWARDS ARROW)	→
& (ampersand)	&	↓ (DOWNWARDS ARROW)	↓
© (copyright)	©	π (Pi)	π
® registered trademark	®	₹ (Rupee)	₹
$\frac{1}{2}$	½	$\frac{1}{4}$	¼
$\frac{2}{3}$	⅔	$\frac{1}{3}$	⅓

Comment in HTML

Comments are text notes added to the program to provide explanatory about the source code.

Syntax:

<!-- Your Comment goes here -->

Example:

<!-- About Section Start here -->

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Image in HTML

- Images can be placed in a web page by using tag.
- The 'src' Attribute is used to set the source of image. The url can be a relative or absolute.
- There are many Image formats available today, but the most widely used among them are .jpg, .jpeg, .png and .gif.

Syntax: ``

Example: ``

Image in a Parent Folder

- project/
- |
- |— images/
- | └─ photo.jpg
- |
- └─ pages/
- └─ about.html

If you want to display photo.jpg inside about.html, use:

Here:

../

means **go one folder back/up**.

Image in HTML (Conti.)

- Attribute of Image tag:
 - Src : to set the source of image (URL/path)
 - Height : to set the height of image (Px or %)
 - Width : to set the width of image (Px or %)
 - Align : to set the alignment of image (Left, Right, center)
 - Border: to define the border of image (Integer Number)
 - Hspace: to set the horizontal space image (Px or %)
 - Vspace: to set the vertical space from image (Px or %)
 - Title : to define the title of image
 - Alt : to set the alternate text for image.

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>Image Attributes Example</title>
5  </head>
6  <body>
7  <h2>HTML Image Attributes Example</h2>
8  <p>
9  This is a sample paragraph to show how the image aligns with text.
10 </p>
11 
21 </body>
22 </html>
```

Link in HTML

<a> ANCHOR TAG:

- Web pages are linked to one another through Hypertext Links.
- Text or image in the HTML document can be linked to an external document or to a specific place within the same document.
- The text or image that provides such linkage is known as Hypertext or Hotspot.

Syntax: Text/Image

Example: <a href="<https://google.com>" />Open google



Link in HTML (Conti.)

- **Attribute of Anchor tag:**

- href : to set the link path/source (url, path,file name)
- name : to specify the name of link as bookmark
- target : to set the target of link (_blank)

- **Link from website:**

- `<a href="https://infomaxacademy.com"> Infomax Academy</a>`

- **Link from other webpage:**

- `<a href="about.html"> About us</a>`

- **Link in same document:**

- `<a name="top"></a>`
- `<a href="#top"> Go to Top</a>`

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Link in Same Document Example</title>
5 </head>
6 <body>
7 <!-- Top of the page -->
8 <a name="top"></a>
9 <h1>Welcome to My Web Page</h1>
10 <p>This is the top section of the page.</p>
11 <hr>
12 <h2>Section 1</h2>
13 <p>
14 This is section 1 content. Lorem ipsum dolor sit amet, consectetur adipiscing elit.
15 </p>
16 <h2>Section 2</h2>
17 <p>
18 This is section 2 content. Scroll down to see the "Go to Top" link.
19 </p>
20 <h2>Section 3</h2>
21 <p>
22 This is section 3 content. More text here to make the page longer.
23 </p>
24 <br><br><br>
25 <!-- Link to go back to top -->
26 <a href="#top">Go to Top</a>
27 </body>
28 </html>
```

Link with Email

- HTML <a> tag provides a option to specify an email address to send an email.
- To link a text with email, use mailto: email_address along with href attribute.

```
<a href="mailto:info@webskynet.com">info@webskynet.com</a>
```

When the user clicks the link, their default email application will open with the recipient email already filled in.

- We can also specify a default email subject and email body along with email address.

```
<a href="mailto:info@webskynet.com?subject=Website  
Inquiry&body=Hello Webskynet IT Solution, I would like to know more  
about your services.">
```

Send Email

```
</a>
```

How It Works

mailto:abc@example.com → Email address where the mail will be sent

subject=Feedback → Pre-filled email subject

body=Message → Pre-filled email message

Send Mail → The clickable text

Link with Mobile/Phone

- To link a text with mobile or phone number, use tel: mobile_number along with href attribute.

`<a href="tel:+23832884"> Call Us</a>`

For WhatsApp : `<a href="https://wa.me/9123832884">Chat on WhatsApp</a>`

Image Linking

Syntax:

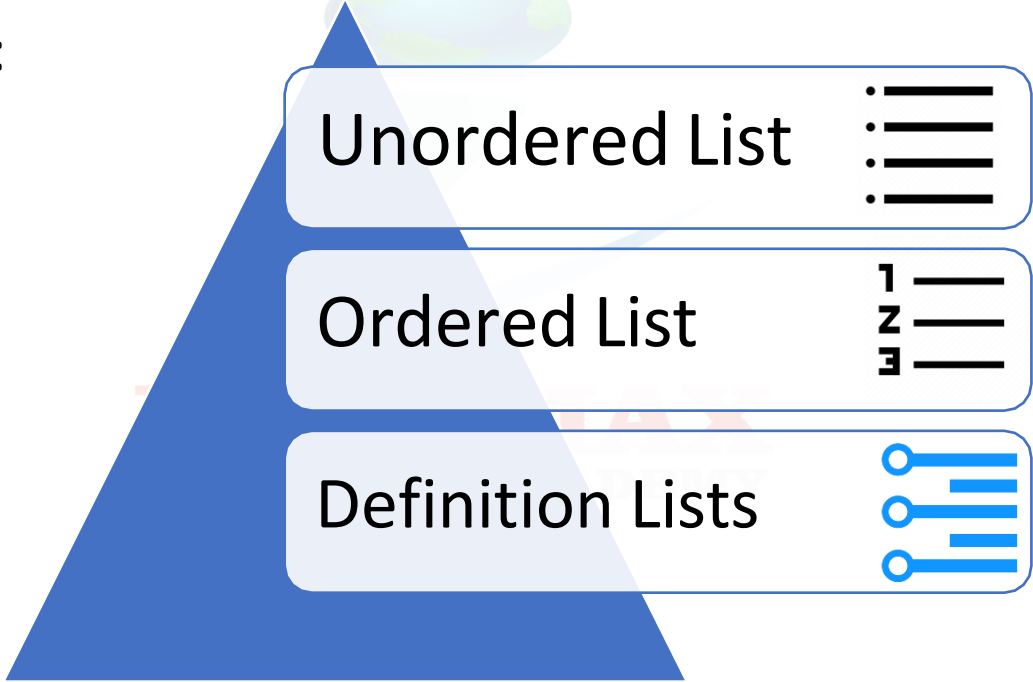
```
<a href="url/path"></a>
```

Example:

```
<a href="http://infomaxacademy.com">  
    
</a>
```

List in HTML:

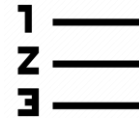
- HTML Supports several ways of arranging items in lists.
- Type of List:



Unordered List



Ordered List



Definition Lists



Unordered List

- Unordered List also called as *bulleted list*, used to present list of items marked with bullets.
- An unordered list starts with in followed by (List Item) tag.

- Syntax:

```
<ul>
```

```
<li>list Item 1</li>
```

```
<li> List item 2</li>
```

```
</ul>
```

Ex- <ul type="square">

```
<li>Item</li>
```

```
<li>Item</li>
```

```
</ul>
```



- Attribute of Tag:

- Type: to set the bullet style.(disk, square, circle) Default – disk



List in HTML : Ordered List

- Ordered list also called as Numbered list, is used to present a numbered list of item in the order of importance or the item (paragraph) is marked with a number.
- An ordered list must begin with the followed by an list item tag.
- Syntax:

```
<ol>
```

```
    <li>list Item 1</li>
```

```
    <li> List item 2</li>
```

```
</ol>
```



Note: can technically work without or , but it's **not valid HTML** and can lead to unexpected behavior.



List in HTML : Ordered List (Conti.)

- Attribute of Tag:
 - **Type** : to define the number style (A,a,I,i,1)
 - **Start** : to set the start value of numbering (any interger)

```
<ol type="A">  
<li>Apple</li>  
<li>Banana</li>  
</ol>
```

```
<ol start="5">  
<li>Item</li>  
<li>Item</li>  
</ol>
```

```
<ol reversed>  
<li>Item</li>  
<li>Item</li>  
</ol>
```

- Attribute of tag:
 - **Value:** to change the track on numbering (any integer)

```
<ol>  
  <li value="5">Item One</li>  
  <li>Item Two</li>  
</ol>
```

- Output starts numbering from **5**

List in HTML : Definition Lists

A definition list is a list of terms and corresponding definitions. Definition lists are typically formatted with the term on the left with the definition following on the right or on the next line. The definition text is typically indented with respect to the term.

The opening list tag must be <DL>. It is followed by an term names (<DT>) and then by definitions (<DD>).

Syntax:

```
<DL>
  <dt>Term 1</dt>
  <dd>This is the definition of the first term.</dd>
  <dt>Term 2</dt>
  <dd>This is the definition of the second term.</dd>
</DL>
```

List in HTML : Nested List

- One list might contain one or more ordered/unordered list that is called as Nested lists.

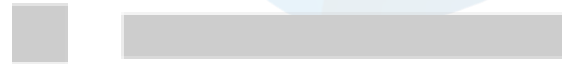
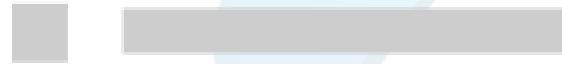


Table in HTML

The **<table>** tag in HTML is used to create tables for displaying data in rows and columns.

It helps organize information clearly, like in spreadsheets.



Create Tables in HTML

Columu 1	Columu 2	Columu 3
Row 1 Cell 1	Row 1 Cell 2	Row 1 Cell 3
	Row 2 Cell 2	Row 2 Cell 3
Row 3 Cell 1		

- Following tag are used to create table:
 - **Table Tag:** The <table> element is used to define a table.
 - **Table Row:** Rows are defined using the <tr> tag (Table Row).
 - **Table Headers:** The <th> tag defines a header cell, typically bold and centered.
 - **Table Data:** Cells in a table are defined using the <td> tag (Table Data).
 - **Table Caption:** The <caption> tag is used to give a title or description to the table. It is placed immediately after the opening <table> tag.
 - **Table Body and Head:**
 - The <thead> tag is used for the table header section.
 - The <tbody> tag is used for the main body of the table.

Example

```
<table border="1">
  <caption>Table 4.5</caption>
  <thead>
    <tr>
      <th>header 1</th>
      <th>header 2</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>row 1, cell 1</td>
      <td>row 1, cell 2</td>
    </tr>
    <tr>
      <td>row 2, cell 1</td>
      <td>row 2, cell 2</td>
    </tr>
  </tbody>
</table>
```

Table in HTML (Conti)

- Attribute of <table> tag:
 - **Height** : to set the height of table
 - **Width** : to set the width of table
 - **Bgcolor** : to set the background color table
 - **Bordercolor** : to set the border color of table
 - **Align** : to set the alignment of table
 - **Cellpadding** : to set the space between text and table border
 - **Cellspacing** : to set the space between two cells
- Attribute of <tr> tag:
 - **Bgcolor** : to set the background color of specified row
 - **Height** : to set the height of specified row

Space inside cell

Cell 1	Cell 2

cellpadding

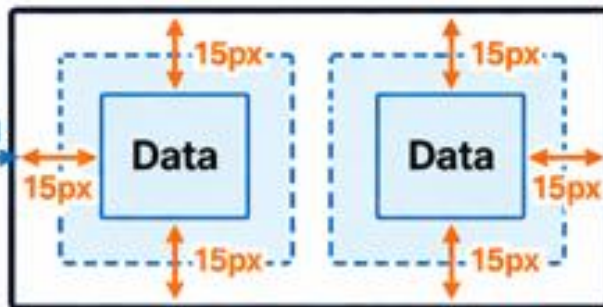
CELLPADDING

Space **inside** the cell
(between content and border)

Example Code:

```
<table border="1" cellpadding="15">
  <tr>
    <td>Data</td>
    <td>Data</td>
  </tr>
</table>
```

Cell Padding
(15px)



Visual Effect with cellpadding="15"

More space inside each cell

Space between cells



cellspacing

CELLSPACING

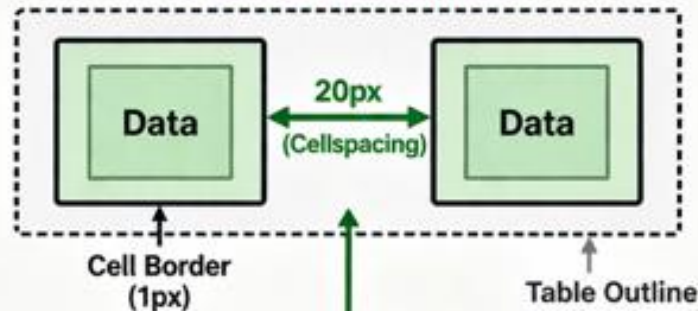
Space **between** cells
(outside the border)

Example Code:

```
<table border="1"
  cellspacing="20">
  <tr>
    <td>Data</td>
    <td>Data</td>
  </tr>
</table>
```

Visual Effect with cellspacing="20"

Space/gap between cells



Property	Affects	Location	Think of as...
cellpadding	Inside the cell	Content ↔ Border	Inner Space
cellspacing	Between cells	Border ↔ Border	Outer Gap



Note: In HTML5, use CSS (padding & border-spacing) instead — attributes are deprecated!

Table in HTML (Conti)

- Attribute of <td> or <th> tag:
 - **Width** : to set the width of specified column
 - **Height**: to set the height of row
 - **Bgcolor**: to set the background color of specified cell
 - **Align** : to set the horizontal alignment of cell text i.e. center, left or right
 - **Valign** : to set the vertical alignment of cell text i.e. top, bottom or center
 - **Rowspan**: to span (merge) the number of rows into one.
 - **Colspan**: to span(merge) the number of column into one.

NOTE : The word “**span**” means **to extend across or cover a certain area**.

1. rowspan – Span Multiple Rows ---

Use **rowspan="number"** → Makes a cell cover **multiple rows** (vertically)

HTML Code:

```
<table border="1">
  <tr>
    <th rowspan="2">Name</th>
    <td>Prerak</td>
  </tr>
  <tr>
    <td>Aarav</td>
  </tr>
</table>
```

↑ 'Name' cell will span 2 rows

Output:

2 Rows Spanned

Name	Prerak
	Aarav

🔑 **rowspan="2"** → "Name" cell covers 2 rows down

2. colspan - Span Multiple Columns

Use `colspan="number"` → Makes a cell cover multiple columns (horizontally)

HTML Code:

```
<table border="1">
  <tr>
    <th>Name</th>
    <th colspan="2">Subjects</th>
  </tr>
  <tr>
    <td>Prerak</td>
    <td>HTML</td>
    <td>CSS</td>
  </tr>
</table>
```

'Subjects' cell will span 2 columns

Output:

2 Columns Spanned

Name	Subjects	
Prerak	HTML	CSS

 `colspan="2"` → "Subjects" cell covers 2 columns across



Quick Reference:

`rowspan="N"`

→ Spans N rows (Vertical ↓)

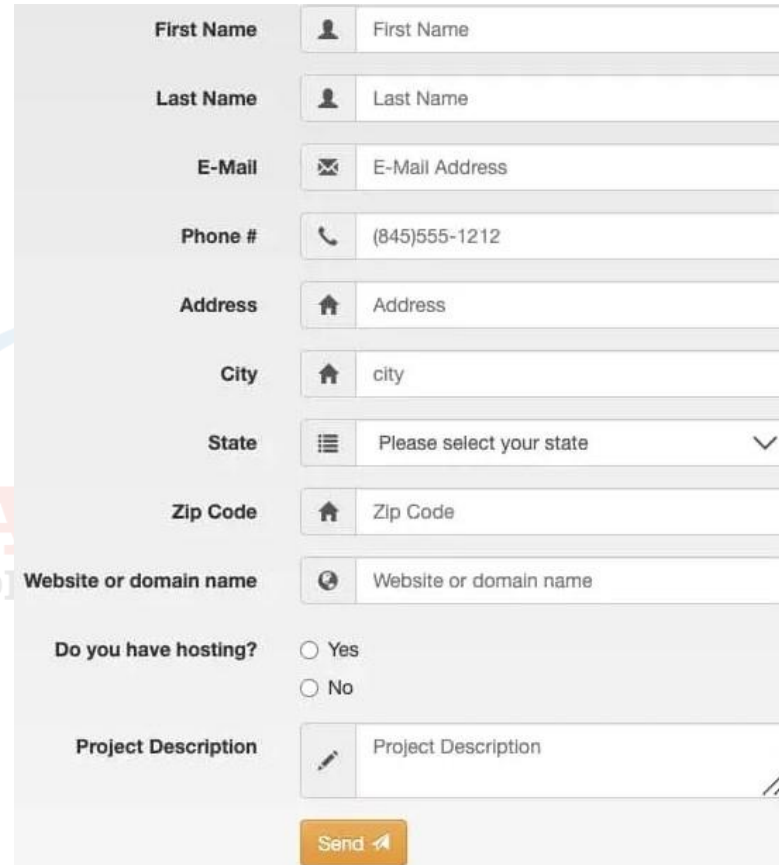
`colspan="N"`

→ Spans N columns (Horizontal →)

Form

An **HTML form** is used to collect user input.
The input can be sent to a server for processing.

- A form can contain input elements like text fields, checkboxes, radio-buttons, submit buttons and more.
- The <form> tag is used to create an HTML form:
 <form>
 Form elements (ex. Textbox, radio button, checkbox etc)
 </form>



A sample HTML form with the following fields and controls:

- First Name**: Text input field with a person icon.
- Last Name**: Text input field with a person icon.
- E-Mail**: Text input field with an envelope icon.
- Phone #**: Text input field with a telephone icon, containing the value "(845)555-1212".
- Address**: Text input field with a house icon.
- City**: Text input field with a house icon, containing the value "city".
- State**: Dropdown menu with a list icon, containing the text "Please select your state" and a downward arrow.
- Zip Code**: Text input field with a house icon.
- Website or domain name**: Text input field with a globe icon.
- Do you have hosting?**: Radio button group with options "Yes" and "No".
- Project Description**: Text area with a pencil icon and a submit button.
- Send**: Orange submit button with a paper plane icon.

HTML Form Input Types & Elements

- **Label:**

Used to display a **text description** for an input field.

```
<label>Student Name</label>
```

- **Text Box:**

Used to accept **single-line text input**.

```
<input type="text" name="user_name">
```

- **Password Field:**

Used to accept **secure data** like passwords or PINs.

Characters are hidden while typing.

```
<input type="password" name="user_pass">
```

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- **Number Input:**

Used to accept **numeric values only**.

You can set limits using min, max, and step.

```
<input type="number" min="0" max="100" step="10">
```

- **Date Input:**

Allows users to select a **date from a calendar**.

```
<input type="date" name="dob">
```

- **Email Input:**

Accepts only **valid email format**.

```
<input type="email" name="user_email">
```

- **Search Box:**
 - Used for search input.
 - Shows a **clear (×) button** inside the box.
 - `<input type="search">`

- **Month Input:**
 - Used to select **month and year only**.
 - `<input type="month">`

- **Time Input:**
 - Used to select **time (hour and minute)**.
 - `<input type="time">`

- **Week Input:**
Used to select **week and year**.
`<input type="week">`

- **Telephone Input:**
Shows a **numeric keypad on mobile devices**.
`<input type="tel">`

- **URL Input:**
Accepts only **valid website URLs** (http/https).
`<input type="url">`



Color Picker:

Allows users to **choose a color**.

```
<input type="color">
```



Range Slider:

Creates a **slider bar** to select a value within a range.

```
<input type="range" min="1" max="1000" step="10">
```



Radio Button:

Used for **multiple options but only one selection**.

```
<input type="radio" name="gender"> Male
```

```
<input type="radio" name="gender"> Female
```

- ❑ **Checkbox:**
Used for **multiple options and multiple selections**.

`<input type="checkbox">` Reading

`<input type="checkbox">` Sports

- ❑ **Textarea:**
Used to accept **multi-line text input**.

`<textarea rows="5" cols="30"></textarea>`

- ❑ **File Upload:**
Creates a **file selection (browse) button**.

`<input type="file">`

- **Select Dropdown:**
Used to create a **dropdown list**.

```
<select>  
  <option>Item 1</option>  
  <option>Item 2</option>  
  <option>Item 3</option>  
</select>
```



Option Group (optgroup):

Used to **group related options** inside a dropdown.

```
<select>
  <optgroup label="Group 1">
    <option>Item 1</option>
    <option>Item 2</option>
  </optgroup>

  <optgroup label="Group 2">
    <option>Item 3</option>
    <option>Item 4</option>
  </optgroup>
</select>
```

Datalist :-

Datalist is used to provide a list of **predefined options** for an input field. When the user starts typing, suggestions appear from the list.

Example –

```
<label>Select Course:</label>  
<input list="courses" type="text" name="course">  
  
<datalist id="courses">  
<option value="CCA">  
<option value="DCA">  
<option value="CCC">  
<option value="Tally">  
<option value="O Level">  
</datalist>
```

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Submit Button:-

A Submit Button is used to submit the form data to the server.

Example –

```
<form>
<label>Name:</label>
<input type="text" name="name">
<input type="submit" value="Save">
</form>
```

Reset Button:-

A **Reset Button** is used to **clear all form fields** and reset them to their default values.

Example –

```
<form>
<label>Name:</label>
<input type="text" name="name">
<input type="reset" value="Clear">
</form>
```

When the user clicks **Clear**, all input fields will be emptied.

Button (Custom Button) :

A Button is used to perform custom actions using JavaScript.

Example –

```
<button type="button" onclick="calculate()">Calculate</button>

<script>
function calculate() {
alert("Calculation Done!");
}
</script>
```

Image Button :

An Image Button is used when you want to use an image instead of a normal button.

Example-

```
<button type="button">

</button>
```

Common Attributes of <input>

1. type

Defines the type of input field (text, password, email, file, etc.).

Example:

```
<input type="text" >
```

2. name

Specifies the name of the input field that is sent to the server when the form is submitted.

Example:

```
<input type="text" name="username">
```

3. value

Defines the default value inside the input field.

Example:

```
<input type="text" value="prerak">
```

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4. id

Provides a unique identifier for the input element. It is mainly used with **labels, CSS, or JavaScript**.

Example:

```
<label for="email">Email:</label>  
<input type="email" id="email">
```

5. placeholder

Displays hint text inside the input field to guide the user.

Example:

```
<input type="text" placeholder="Enter your full name">
```

6. required

Makes the input field mandatory before submitting the form.

Example:

```
<input type="email" placeholder="Enter email" required>
```

7. readonly

Makes the field non-editable, but its value **will still be submitted** with the form.

Example:

```
<input type="text" value="India" readonly>
```

8. disabled

Makes the field non-editable and its value **will NOT be sent to the server**.

Example:

```
<input type="text" value="Admin" disabled>
```

Common Attributes of <input>

1. maxlength

Defines the **maximum number of characters** a user can enter in an input field.

Example:

```
<input type="text" maxlength="10" placeholder="Enter username">
```

The user can type **only up to 10 characters**.

2. min / max

Specifies the **minimum and maximum values** for inputs like **number, date, range, etc.**

Example:

```
<input type="number" min="1" max="10">
```

The user can enter numbers **between 1 and 10 only**.

Date Example:

```
<input type="date" min="2024-01-01" max="2026-12-31">
```

3. step

Defines the **interval between allowed values** for number or date inputs.

Example:

```
<input type="number" min="0" max="100" step="5">
```

Allowed values: **0, 5, 10, 15, 20, ...**

4. size

Specifies the **visible width of the input field in characters**.

Example:

```
<input type="text" size="30">
```

The input box will be **30 characters wide**.

5. autofocus

Automatically places the **cursor inside the input field when the page loads**.

Example:

```
<input type="text" placeholder="Enter your name" autofocus>
```

6. checked

Used with **checkbox or radio buttons** to make them **selected by default**.

Example (Checkbox):

```
<input type="checkbox" checked> Subscribe Newsletter
```

Example (Radio):

```
<input type="radio" name="gender" checked> Male
```

```
<input type="radio" name="gender"> Female
```

7. accept

Specifies the **type of files allowed for upload** in a file input.

Example:

```
<input type="file" accept=".jpg,.png,.pdf">
```

Only **JPG, PNG, and PDF** files can be selected.

Example Form Using These Attributes

<form>

Username: <input type="text" maxlength="12">

Age: <input type="number" min="18" max="60">

Marks: <input type="number" step="5">

Search: <input type="text" size="30">

Email: <input type="email" autofocus>

Subscribe: <input type="checkbox" checked>

Upload Image: <input type="file" accept=".jpg,.png">

<input type="submit">

</form>

HTML contenteditable global attribute

The **contenteditable** global attribute is an enumerated attribute indicating if the element should be editable by the user. If so, the browser modifies its widget to allow editing.

The attribute must take one of the following values:

- true or an *empty string*, which indicates that the element is editable.
- false, which indicates that the element is not editable.

```
<blockquote contenteditable="true">
```

```
<p>Edit this content to add your own uote</p>
```

```
</blockquote>
```

```
<cite contenteditable="true">-- Write your own name here</cite>
```

Field set and Legend

Fieldset :

The **<fieldset>** tag is used to **group related elements in a form**. It draws a **box around the grouped form controls**, which makes the form more organized and easier to understand.

```
<form>
```

```
<fieldset>
```

```
<label>First Name:</label>
```

```
<input type="text"><br><br>
```

```
<label>Last Name:</label>
```

```
<input type="text">
```

```
</fieldset>
```

```
</form>
```

Note : The input fields will appear inside a bordered box.

Legend:

The **<legend>** tag is used to **define a caption or title for the <fieldset> element**. It describes the group of form elements inside the fieldset.

```
<form>
```

```
<fieldset>
```

```
<legend>Student Information</legend>
```

```
<label>Name:</label>
```

```
<input type="text"><br><br>
```

```
<label>Email:</label>
```

```
<input type="email"><br><br>
```

```
<label>Course:</label>
```

```
<input type="text">
```

```
</fieldset>
```

```
</form>
```

Frameset

- The **<frameset>** tag in HTML was used to **divide the browser window into multiple sections (frames)**, where each frame can load a **separate HTML document**.

Note: The **<frameset>** tag is **deprecated in HTML5** and is no longer used in modern web development. Today developers use **CSS, Flexbox, Grid, or iframes** instead.

A **frameset** replaces the **<body>** tag and defines how the screen is divided.

```
<!DOCTYPE html>
<html>
<head><title>Frameset Column Example</title></head>
<frameset cols="30%,70%">
  <frame src="menu.html">
  <frame src="content.html">
</frameset>
</html>
```

```
<frameset rows="size,size">
<frame src="page1.html">
<frame src="page2.html">
</frameset>
```

Frameset (Conti.)

- The **<frame> tag** is used inside the **<frameset>** tag to display another HTML document within a frame.

Each frame can have different attributes to control its behavior.

1. name Attribute

The **name attribute** is used to **give a name to the frame**. This name can be used as a **target for links**.

Example

```
<frameset cols="30%,70%">  
<frame src="menu.html" name="menu">  
<frame src="content.html" name="content">  
</frameset>
```

2. src Attribute

The **src attribute** specifies the **HTML file** that will be displayed inside the frame.

Example

```
<frameset cols="50%,50%">  
<frame src="page1.html">  
<frame src="page2.html">  
</frameset>
```

3. border Attribute

The **border attribute** is used to **define the thickness of the frame border**.

Example

```
<frameset cols="50%,50%" border="5">  
<frame src="left.html">  
<frame src="right.html">  
</frameset>
```

4. noresize Attribute

The **noresize attribute** prevents the user from **resizing the frame**.

Example

```
<frameset cols="30%,70%">  
<frame src="menu.html" noresize>  
<frame src="content.html">  
</frameset>
```

5. scrolling Attribute

The **scrolling attribute** controls whether **scrollbars appear in the frame**.

Values:

- yes → Scrollbar always visible
- no → Scrollbar hidden
- auto → Scrollbar appears only when needed

Example

```
<frameset cols="50%,50%">  
<frame src="page1.html" scrolling="yes">  
<frame src="page2.html" scrolling="no">  
</frameset>
```

IFrame

- The **<iframe> (Inline Frame)** tag is used to **embed another webpage or document inside the current HTML page.**
- Unlike <frameset>, an iframe is placed **inside the <body> tag.**
- The Document may contain text, video, map etc.
- `<iframe src="url" height="" width="" ></iframe>`
- Others Attribute of <iframe>
 - frameborder (yes, no)
 - Scrolling (yes, No , Auto)
 - name

Example with Multiple Attributes

```
<!DOCTYPE html>
<html>
<body>
<h2>HTML IFrame Example</h2>
```

```
<iframe
src="https://www.example.com"
width="500"
height="300"
name="myframe"
frameborder="1"
scrolling="auto">
</iframe>
</body>
</html>
```

Link Targeting an IFrame

```
<!DOCTYPE html>
<html>
<body>

<a href="page2.html"
target="frame1">Open Page</a>

<br><br>

<iframe name="frame1" width="500"
height="300"></iframe>

</body>
</html>
```

Difference between Frame and IFrame

Frame

Used with `<frameset>`

Divides the whole browser window

Deprecated in HTML5

IFrame

Used inside `<body>`

Embeds a page inside another page

Still widely used

Audio in HTML

- The **<audio>** tag in HTML5 is used to **embed sound files** (music, speech, etc.) in a web page. Users can play, pause, and control the audio.

Common Attributes

Attribute

controls

autoplay

loop

muted

Description

Displays play/pause controls

Audio starts automatically

Audio repeats continuously

Audio starts in muted mode

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>Audio Example</h2>
```

```
<audio controls>
```

```
  <source src="music.mp3" type="audio/mpeg">
```

```
  Your browser does not support the audio element.
```

```
</audio>
```

```
</body>
```

```
</html>
```

Video in HTML

- The HTML <video> element is used to show a video on a web page.

Common Attributes

Attribute

controls

width

height

autoplay

loop

muted

Description

Shows video controls

Sets video width

Sets video height

Video starts automatically

Video repeats continuously

Video starts muted

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>Video Example</h2>
```

```
<video width="400" height="300" controls>
```

```
  <source src="movie.mp4" type="video/mp4">
```

```
  Your browser does not support the video tag.
```

```
</video>
```

```
</body>
```

```
</html>
```

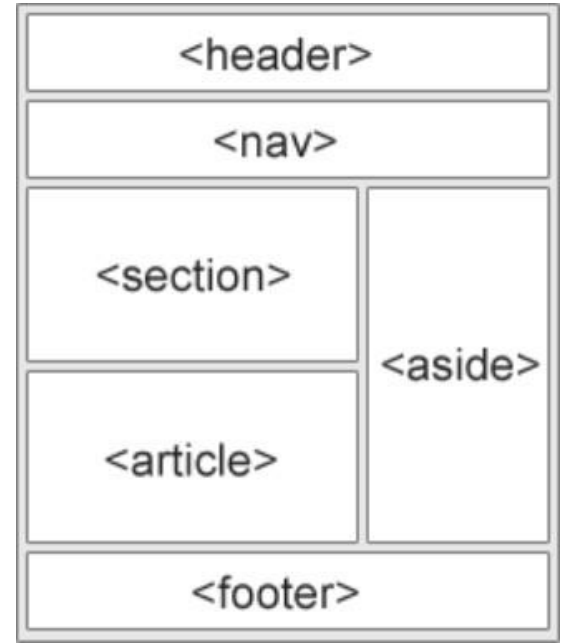
HTML 5 Tags

- HTML5 introduced many **new semantic tags** that help organize web pages and improve SEO.

Common HTML5 Tags

Tag	Purpose
<header>	Defines header section
<nav>	Defines navigation links
<section>	Defines a section in a page
<article>	Defines independent content
<aside>	Defines sidebar content
<footer>	Defines footer section
<figure>	Used for images with caption
<figcaption>	Caption for figure

```
<!DOCTYPE html>
<html>
<body>
<header>
<h1>My Website</h1>
</header>
<nav>
<a href="#">Home</a> |
<a href="#">About</a> </nav>
<section>
<article>
<h2>HTML5 Introduction</h2>
<p>HTML5 provides new semantic elements.</p> </article>
</section>
<footer><p>Copyright 2026</p>
</footer>
</body>
</html>
```



<figure> Tag in HTML :

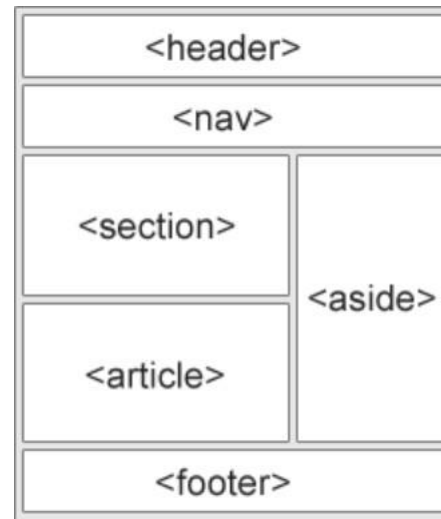
The **<figure>** tag in HTML5 is used to **group media content** such as images, diagrams, illustrations, code snippets, or charts.

It is often used together with the **<figcaption>** tag to provide a **caption or description** for the content.

```
<!DOCTYPE html>
<html>
<body>
<h2>Figure Tag Example</h2>
<figure>
  
  <figcaption>Fig.1 - A Desktop Computer</figcaption>
</figure>
</body>
</html>
```

HTML 5: Sectioning Elements:

- <header>
- <footer>
- <nav>
- <article>
- <section>
- <aside>



HTML 5 : Multimedia Elements:

- <audio>
- <video>
- <source>
- <track>

- <canvas>
- <svg>



HTML 5 : Interactive Elements:

- <details>
- <summary>
- <mark>
- <progress>
- <meter>



HTML 5 : Form Elements:

- <datalist>
- <output>
- <input> (range,color,date,etc)
- <textarea>
- <button>
- <select>
- <select>
- <optgroup>
- <option>
- <fieldset>
- <legend>
- <label>

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HTML 5 : Semantic Elements:

- `<main>` : `<main> <h1>Welcome to My Website</h1></main>`
- `<figure>` : `<figure></figure>`
- `<figcaption>` : `<figcaption>Beautiful nature view</figcaption>`
- `<mark>` : `<p>This is <mark>important</mark> text.</p>`
- `<progress>` : `<progress value="70" max="100"></progress>`
- `<output>` : `<output name="result"></output>`
- `<time>` : `<p>Event date: <time datetime="2026-04-15">April 15, 2026</time></p>`
- `<ruby>`
- `<rt>`
- `<rp>`

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HTML 5 Semantic element for Text Formatting

- `<abbr>`: Represents an abbreviation or acronym.

`<abbr title="World Health Organization">WHO</abbr>`

- `<em>`: Represents text with emphasis.

This is `<em>very</em>` important.

- `<cite>`: Represents the title of a creative work (e.g., book, movie, song) or the name of its author.

`<cite>The Great Gatsby</cite>` by F. Scott Fitzgerald

Conti.

- `<pre>`: Represents preformatted text, preserving whitespace and line breaks. Ex.

```
<pre>
Infomax Computer Academy
  G. R. Complex Preetam Nagar
Dhoomanganj, Uttar Pradesh
  India
</pre>
```

- `<q>`: Represents inline quotations, typically enclosed within quotation marks.
She said, `<q>I'll be there soon</q>`.

Conti.

- `<blockquote>`: Represents a block of quoted text, often used for longer quotations.

`<blockquote>"The only way to do great work is to love what you do." - Steve Jobs </blockquote>`

- `<dfn>`: Represents a definition term, typically rendered in italics by default.

`<dfn>HTML</dfn>` stands for HyperText Markup Language.

- `<time>`: Represents a specific time or date, optionally with a machine-readable format.

Join us on `<time datetime="2024-03-19">March 19th</time>` for the event.

Conti.

- `<var>`: Represents a variable or placeholder, typically rendered in italics by default.

The formula is `<var>y = mx + b</var>`.

- `<kbd>`: Represents keyboard input, typically displayed in a monospaced font.

To save, press `<kbd>Ctrl + S</kbd>`.

- `<samp>`: Represents sample output from a computer program.

The output of the command is `<samp>hello world</samp>`

- `<code>`: Represents a fragment of computer code, typically displayed in a monospaced font.
Use the `<code>print()</code>` function to display output.
- `<span>`: Provides a generic container for inline elements.
- `<div>`: Provides a generic container for block-level elements.

Thank You

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